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News Shield: An Intelligent System for Detection and Verifying Fake News

PROJECT SYNOPSIS

OF MAJOR PROJECT

BACHELOR OF TECHNOLOGY

COMPUTER SCIENCE & ENGINEERING

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NOTATIONS / NOMENCLATURE

ML - Machine Learning

NLP - Natural Language Processing

TF-IDF - Term Frequency-Inverse Document Frequency

SVM - Support Vector Machine

LSTM - Long Short-Term Memory

BERT - Bidirectional Encoder Representations from Transformers

1. INTRODUCTION

Background and problem statement

Digital communication allows news and claims to spread quickly through social media, websites, and messaging applications. Alongside legitimate reporting, fabricated, distorted, or misleading material may also be presented as news. Readers may have difficulty checking every item, while manual fact-checking takes time. The project addresses this need by proposing a quick screening tool for news-style text.

The proposed News Shield system will use supervised machine learning to classify a news article, headline, or social-media post into dataset-defined “real” or “fake” categories. It will learn patterns from labelled examples. The model output is an estimate based on those examples; it does not independently verify evidence, confirm the source, or establish that a claim is true.

Proposed approach and technology

The text-processing pipeline may include cleaning, tokenization, stop-word handling, and stemming or lemmatization. Candidate features include TF-IDF, Bag-of-Words, or word embeddings such as Word2Vec or GloVe. The proposed baseline comparison is Logistic Regression, Naive Bayes, Support Vector Machine, and Random Forest. If time and computing resources permit, LSTM/Bi-LSTM and BERT-based models may be explored as advanced comparisons.

The project proposes an interactive web application in which a user can enter a headline or article and receive a “Real” or “Fake” prediction with a model confidence score. Flask or Django may be used for the application, subject to the team’s implementation choice. Any score will be labelled as a model estimate, not a verified probability of truth. The system should help readers decide what to check; important claims still require original and independent sources.

Scope and expected contribution

The initial scope is text classification using a documented labelled dataset. Social-context signals, source-credibility indicators, transformer models, and deployment are possible extensions and will be included only if they can be implemented and evaluated within the project scope. Expected deliverables are a documented data pipeline, a model comparison using held-out data, an interactive prototype, and a report of errors and limitations. No accuracy level is assumed in advance.

2. LITERATURE REVIEW

Shu et al. survey fake-news detection from a data-mining perspective and organize prior work around news content, social context, datasets, and evaluation [1]. Their review highlights that text is only one possible signal. This supports the proposed text-focused baseline while motivating a clear statement that it does not use all available context.

Pérez-Rosas et al. introduced datasets covering multiple online news domains and evaluated linguistic approaches for automatic detection [2]. Ahmed, Traoré, and Saad studied n-gram

analysis with machine-learning classifiers [3]. These works motivate testing conventional text features and classifiers, but their results belong to their specific datasets and experimental settings; they do not guarantee the same performance for this project.

Wang's LIAR benchmark contains manually labelled short political statements and provides a benchmark for fine-grained classification [4]. Ruchansky et al.'s CSI model combines article text with user-response and source-behaviour information [5]. These works show the difference between a text-only prototype and systems that incorporate context or social signals.

Kaliyar et al. proposed FakeBERT, combining BERT representations with convolutional layers for social-media fake-news classification [6]. Devlin et al. introduced BERT as a bidirectional language representation model [8]. FakeNewsNet provides news content together with social and spatiotemporal information for research [7]. The team's supplied draft also identifies the Kaggle Fake and Real News dataset and PolitiFact as possible data or fact-checking resources [9], [10]. Availability, licensing, labels, and suitability must be checked before choosing a dataset.

The reviewed work ranges from conventional n-gram models to contextual deep-learning and social-context methods. The proposed mini project will first establish a reproducible baseline, then compare selected alternatives if resources permit. Evaluation on held-out examples, error analysis, and human verification are necessary because a classifier can fail on new domains, satire, or changing language.

3. OBJECTIVES

1. Study existing fake-news detection approaches, datasets, and techniques relevant to the selected project scope.
2. Select and document a labelled real/fake news dataset, including its provenance, licence, categories, and limitations.
3. Pre-process the text and compare suitable feature representations such as TF-IDF and Bag-of-Words; treat embeddings as an optional extension.
4. Train and evaluate baseline classifiers, including Naive Bayes, Logistic Regression, and SVM; add other models only when feasible.
5. Explore an LSTM/Bi-LSTM or BERT-based model as an optional advanced comparison if time and compute allow.
6. Compare models with accuracy, precision, recall, F1-score, and a confusion matrix on held-out data; use cross-validation only within training data when appropriate.
7. Develop a user-friendly web interface for text input and a cautious prediction with a clearly explained model confidence score.
8. Review errors and describe limits in generalizing across topics, domains, languages, and writing styles.

4. METHODOLOGY / PLANNING OF WORK

1. Literature and requirements: confirm the project scope, labels, data permissions, interface needs, and evaluation plan with the guide.
2. Dataset selection: review candidate resources such as LIAR, FakeNewsNet, or the supplied draft's Kaggle dataset; record provenance, licence, label definitions, and known limitations before selecting one.
3. Data preparation: inspect class balance, missing values, and duplicates; split data into training and test sets before fitting preprocessing or feature transformations.
4. Text processing and features: apply consistent cleaning, tokenization, and any selected stop-word or stemming/lemmatization steps. Compare TF-IDF and Bag-of-Words first; embeddings are optional.
5. Model training: establish baseline models such as Naive Bayes, Logistic Regression, and SVM. Consider Random Forest, LSTM/Bi-LSTM, or BERT only as staged comparisons when feasible.
6. Evaluation: report accuracy, precision, recall, F1-score, and a confusion matrix on held-out data; inspect false positives and false negatives. Do not report results before the experiment is run.
7. Application development: build a simple front end and Flask or Django back end, integrate the selected preprocessing and trained model, and show a prediction with a caution about model uncertainty.
8. Testing and documentation: test unseen examples, document limitations and future extensions, and prepare the final report. The prototype is a screening aid, not an automated authority on truth.

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